Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

Izhar Ali [1]

Abstract

When language models disagree on a question, the disagreement is often read as missing knowledge. Questions that need the same missing knowledge should then be answered right or wrong together. We look for this structure in the eigenvalues of the correlation matrix of the questions. Within one model, answers drawn independently show no structure the test can detect. Structure appears when all questions in one pass over the benchmark share one random seed or one few-shot prompt, drawn anew for each pass. Across 24 models, the largest structure beyond overall skill is a preference among answer letters. Without one model, Llama-3.2-1B, that eigenvalue falls within the range of one skill, though a weaker letter grouping remains. When the options are rotated, that structure follows the letter far more than the question. Counting a question correct only if it is right in every rotation removes this structure, at least when the preference acts only on the model's guesses. A marginally significant, fragile excess remains, tied post hoc to two pairs of near-duplicate models.

1. Introduction

Sampling-based methods such as self-consistency, SelfCheckGPT and semantic entropy sample a language model several times on the same question and read disagreement among the answers as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). In a deep ensemble, disagreement among the members is read as epistemic uncertainty: the part that comes from missing knowledge, rather than from noise that would remain even if the model were known (Lakshminarayanan et al., 2017; Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Recent scores add disagreement between models to self-consistency and treat it as epistemic (Hamidieh et al., 2026).

This reading makes a prediction. A model that lacks some piece of knowledge should fail every question that needs it, so across an ensemble those questions should be right or wrong together. We call that tendency *cross-question structure*. Related structure is already documented: different models often choose the same wrong answer (Kim et al., 2025), and a few latent factors summarize benchmark scores (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). We ask whether cross-question structure really comes from shared knowledge.

[1]Rowan University. Correspondence to: Izhar Ali <aliz@rowan.edu>.

2nd Workshop on Epistemic Intelligence in Machine Learning (EIML@ICML 2026), Seoul, South Korea. *Copyright 2026 by the author(s).*

Two other sources can produce the same structure, and both are easy to miss. The first is the evaluation pipeline. Within one model, knowledge is fixed, so each question has the same chance of a correct answer on every pass (pass k is the model's kth answer to every question, not pass@k). Knowledge therefore cannot make two questions right on the same passes and wrong on the others; only something the questions share inside a pass can. A shared random seed is one example. All questions in the pass then receive the same random numbers from the sampler, so a draw that favors the letter "B" at one generation step favors it in every question whose answer letter is emitted at that step. A shared few-shot prompt is another, since few-shot examples shape answers (Zhao et al., 2021; Min et al., 2022).

The second source is a letter preference of the model. Llama-3.2-1B is accurate when the correct answer is placed at "A", and below the 0.25 of a random guess when it is placed at the other letters (Fig. 1). It answers "A" in 81% of its replies in the original option order, and a model fitted to its answers puts 0.90 of its guesses on "A" (App. F). Questions whose correct answer is "A" then look like questions it knows, and across an ensemble those questions are right or wrong together. That is the pattern shared missing knowledge would produce. Letter preferences of single models (also called selection bias) are already documented (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024; Ding, 2026). Rotating the options separates a letter preference from a knowledge gap. A knowledge gap stays with the question. A letter preference stays with the answer's letter.

We test the two sources separately. Within one model, we

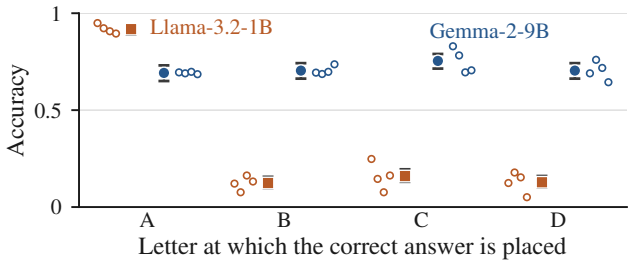


Figure 1. Llama-3.2-1B is accurate where the answer is placed at "A", in each of the four rotations; Gemma-2-9B is about equally accurate at every letter. Filled: all 500 questions, with the options rotated so that the correct answer is at that letter; bars: Wilson 95% CI. Open: each of the four rotations alone (about 125 questions).

compare independent draws with passes that share a seed or a few-shot prompt (Sec. 3.1). Across 24 models, we rotate the options to new letters, so the gold letter (the letter of the correct option) changes and the question and options do not (Sec. 3.2), and we count a question correct only when the model is right in every rotation (*all-4-correct*; Sec. 3.3). All three tests use the *correctness matrix*: 1 if an answer is right, 0 if wrong. Structure is an eigenvalue of the correlation matrix of its questions that is larger than a null model allows: independent answers within one model, and, across models, one skill per model (a unidimensional item response model). Spectral summaries of benchmark dimensionality exist (Sha & Zhao, 2026). We add that the correctness matrix records no letters, yet the second eigenvector of its correlation matrix follows a letter preference, and removing one model brings its eigenvalue within the range of one skill (Sec. 3.2).

2. A spectral test for cross-question structure

We record the answers in a correctness matrix $\mathbf{C}$. Entry c_{ki} is 1 if row k answers question i correctly and 0 otherwise. Within one model a row is a pass; across models a row is a model. We keep only questions that vary: within a model, the N_b *borderline* questions, whose pass rate f_i (the share of passes that answer question i correctly) lies between $\varepsilon = 0.05$ and $1 - \varepsilon$; across models, the *split* questions, which some models answer correctly and others do not.

Let $\mathbf{R}$ be the matrix of Pearson (phi) correlations between the 0/1 columns of the kept questions. If the questions were independent, every true correlation would be 0 and every eigenvalue would be 1. With a finite number of rows, the measured correlations are noisy, and the noise spreads the eigenvalues around 1. Marchenko and Pastur showed that the eigenvalues then fill a range whose upper edge is $\lambda_+ = (1 + \sqrt{\gamma})^2$, where γ is the number of questions divided by the number of rows (Marčenko & Pastur, 1967); as both numbers grow, the largest eigenvalue of such a matrix of independent entries converges to that edge (Bai & Silverstein, 2010; Jiang, 2004). More questions per row give a higher edge. For Llama-3-8B's 173 borderline questions and 100 passes, $\gamma = 1.73$ and $\lambda_+ = 5.36$. An eigenvalue above that edge is evidence that some group of questions is right or wrong together more often than noise allows. Its eigenvector gives each question's *loading*, how strongly it belongs to the group, and each row's *score*, its projection.

Within one model, knowledge is fixed, so only the pipeline can make questions right or wrong together (Sec. 1). This test checks the pipeline, not knowledge, and its null model is independence. We count the eigenvalues above the Marchenko–Pastur (MP) edge λ_+ (the *MP count*). Because λ_+ is a limit for large matrices, we also compare the largest eigenvalue with its values in 1000 simulated matrices of the same size, in which each question keeps its observed pass rate and every answer is drawn independently (App. C).

Across models, independence is the wrong null model. A strong model is right on many questions and a weak one on few, so any two questions are correlated through skill alone; this gives one large eigenvalue, λ_1. The null model is a one-skill item response model. Model m has skill θ_m, question i has difficulty b_i and discrimination (slope) a_i, and the probability of a correct answer is $\sigma(a_i(\theta_m - b_i))$, with σ the logistic function. This is the two-parameter logistic (2PL) model (Lord & Novick, 1968). With 0/1 answers, skill alone can also produce a second large eigenvalue, the *difficulty factor* of binary items (Ferguson, 1941; McDonald & Ahlawat, 1974), because a 0/1 question correlates more strongly with questions of similar difficulty. In the all-4-correct matrix of Sec. 3.3 (391 split questions, 24 models), $\lambda_+ = 25.4$, yet one-skill simulations have a median λ_2 of 28.2. We fit the model, simulate datasets from it, and say that λ_j, the jth largest eigenvalue (rank j), *exceeds* the null when it lies above the 99th percentile of its simulations.

Two nulls ignore guessing: the 2PL, and Rasch, which gives every question the same discrimination (Rasch, 1960). Because a model that does not know the answer is still right a quarter of the time, we also fit the three-parameter logistic (3PL), with its guessing parameter, the lowest chance of a correct answer, fixed at 0.25, and knowledge-or-random-guessing (*know-or-guess*) nulls. In the uniform one, KG-u, a model that does not know the answer guesses one of the four letters with equal probability (App. F). A null is *adequate* when the observed λ_1, the skill eigenvalue, falls between that null's 1st and 99th percentiles; a null that cannot reproduce skill cannot judge what lies beyond it.

Structure beyond skill can appear in λ_2, λ_3 or λ_4. The *max-statistic test* tests all three at once: it takes the largest of their percentiles among simulated eigenvalues of the same rank and compares it with the same maximum in each simulated dataset. Its p-value is p_M.

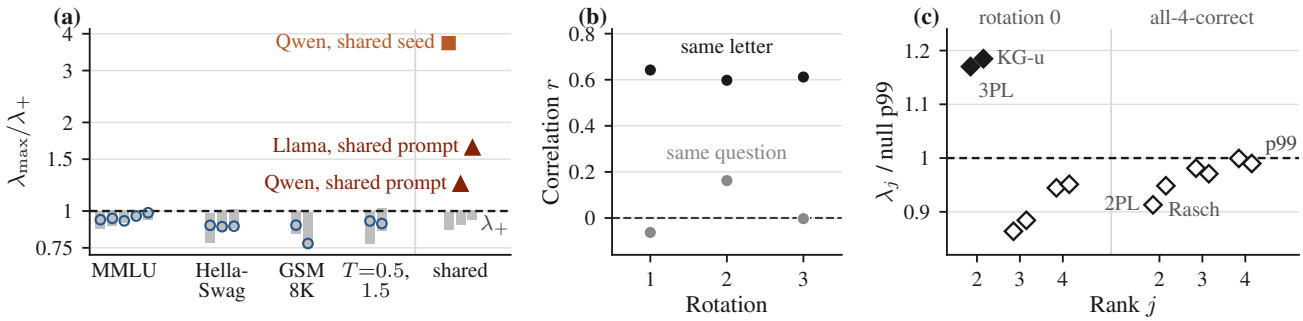


Figure 2. **(a)** Within one model: $\lambda_{\max}/\lambda_+$ (log axis) of each run; grey bar: 5–95% of its simulations; circles: independent draws; filled: an eigenvalue above λ_+. **(b)** Across models, rotations 1–3: correlation over questions of each question's eigenvector-2 loading with the mean rotation-0 loading of the questions that had its new gold letter (same letter), or with its own rotation-0 loading (same question). **(c)** λ_2–λ_4 of the 24-model matrix in rotation 0 and all-4-correct, divided by the 99th percentile (1000 draws) of two adequate one-skill nulls (rotation 0: 3PL left and KG-u, uniform know-or-guess, right; all-4-correct: 2PL left, Rasch right); filled: above it. Other nulls: App. F (soft-preference: not rejected).

Rotations. Each model answers every question four times, with the options moved zero (rotation 0, the original order), one, two or three letters later, cyclically. We count a question correct only if it is right in all four rotations: all-4-correct, the CircularEval rule of Liu et al. (2024). A guess with any fixed letter preference is then correct with probability at most $1/256$ on every question, because the correct answer visits each letter once and the product of the four letter probabilities is at most $(1/4)^4$.

Runs. Within one model there are twelve *independent-draw runs* on MMLU (Hendrycks et al., 2021), HellaSwag (Zellers et al., 2019) and GSM8K (Cobbe et al., 2021), in which each answer has its own seed, as in self-consistency (Table A1). Each run has $K = 30$ to 100 passes. One *shared-seed run* gives all questions of a pass one seed, and two *shared-prompt runs* draw five MMLU development questions (the few-shot *exemplars*) for each pass and place them, with their answers, before every question of that pass. Across models, 24 models from ten families answer the 500 MMLU questions in all four rotations by greedy decoding. A reply that cannot be read as a single answer is *unparseable* and scored wrong; across models, 217 of the 500 questions have at least one such reply (App. B).

3. Results

3.1. Within one model: detectable structure only from a shared seed or prompt

In all 12 independent-draw runs, the MP count is zero: the largest eigenvalue stays below λ_+ and between the 5th and 95th percentiles of its simulations (Fig. 2a; App. C). The test detects weak structure spread over many questions: a factor shared by the questions of a pass, with latent correlation 0.05 (0/1 correlation about 0.025), is detected in every simulation at Llama-3-8B's 173 borderline questions, and in 47% of simulations at Qwen2.5-7B's 43. A factor on only nine questions, by contrast, even at latent correlation 0.2 and picked at random or from one subject, is detected in at most a quarter of simulations (11 to 24%; App. D).

Sharing a seed or a prompt changes this. In all three such runs the largest eigenvalue lies far above every simulation, and the MP count is two to four. For the shared seed, the cause is the sampler's shared random numbers (Sec. 1; App. E). In Qwen2.5-7B's shared-seed run, the gold letter explains 73% of the variance of the top eigenvector's loadings ($\eta^2 = 0.73$, permutation $p = 0.0002$). With a shared prompt, Llama-3-8B copies the exemplars' answer letters, and its top eigenvector also groups by gold letter ($\eta^2 = 0.53$). Qwen2.5-7B's top eigenvector does not group by gold letter. Its passes are coupled by the exemplars themselves: passes that share an exemplar agree more on borderline questions than passes that share none ($p = 0.0005$).

3.2. Across models: the second eigenvector follows the answer letter

In rotation 0, the first eigenvector of the 24-model matrix is skill: the models' scores on it correlate with their accuracy at $r = 0.94$. Three nulls with guessing are adequate: the 3PL, KG-u, and the letter-aware null, a know-or-guess null in which a model that is guessing uses its own fitted letter preference. The second eigenvalue, 49.9 on 460 split questions, exceeds the 99th percentile of all three, and no other eigenvalue does. The max-statistic test rejects them ($p_M = 0.002$; Fig. 2c).

The loadings on this second eigenvector group by gold letter, and the models' scores on it follow their share of "A" answers ($r = 0.76$) and, with the opposite sign, their share of "D" answers ($r = -0.90$). Within one rotation each question's gold letter is fixed, so letter and content cannot be told apart; across rotations they can. Suppose a question's

correct answer moves from "B" in rotation 0 to "C" in rotation 1. If its loading comes from content, its new loading matches its own rotation-0 loading. If it comes from a letter preference, its new loading matches the average rotation-0 loading of the questions whose answer was at "C". The eigenvector follows the letter ($r = 0.60$ to 0.64) and barely follows the question ($r = -0.06$ to 0.16; Fig. 2b).

A fourth adequate null, the *soft-preference* null, puts a fitted letter preference on top of one skill and lets it act on every answer, not only on guesses. The max-statistic test does not reject this null ($p_M = 0.079$), although the null fails its predictive check. A check of λ_2 on its own still finds the observed value unlikely ($p = 0.010$; App. F). The fitted letter preference explains part of the eigenvector, not all of it.

The eigenvalue depends on one model, Llama-3.2-1B. Removing it lowers λ_2 from 49.9 to 35.0, below the 99th percentile of the 3PL null on the other 23 models; removing any other single model leaves it between 47.1 and 52.4. Without Llama-3.2-1B the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$), and the max-statistic test still rejects the 3PL null through λ_4 ($p_M = 0.003$; App. F).

3.3. After rotation: a marginal excess that depends on two model pairs

All-4-correct cancels a letter preference that acts only on guesses. In the all-4-correct matrix (391 split questions), λ_2 lies below the 99th percentile of both adequate one-skill nulls (Fig. 2c). The difficulty factor makes those null distributions wide, and a real second skill could hide inside them. We planted one, as strong as the first, on the 125 kept questions from STEM subjects. The max-statistic test detected this planted skill in more than 90% of datasets, and one half as strong in about a fifth (App. F).

What remains is a marginal excess on λ_3 and λ_4. Only λ_4 under the 2PL exceeds its 99th percentile, at the 99.1st in 2000 draws (Table A2; just below it in the 1000 of Fig. 2c), and no eigenvalue exceeds both nulls. The max-statistic test still rejects both ($p_M = 0.026$ for the 2PL and 0.041 for Rasch). Under a true one-skill model the same test rejects slightly too often, in up to 8.7% of datasets at the 5% level; recalibrated against datasets simulated from the fitted models, p_M is 0.033 and 0.047 for the 2PL test and 0.057 and 0.017 for the Rasch test. Dropping the 217 questions that contain an unparseable reply, which leaves 236 split questions, ends the rejection.

The excess depends on near-duplicate models. Before running the max-statistic test, we listed the *sibling groups*, successive releases or fine-tunes in one family at one size: Phi-3/3.5-mini, Llama-3/3.1-8B, Qwen2/2.5-7B, and Mistral-7B v0.2/v0.3 with Zephyr-7B. Keeping one model from each group, in any of the 24 ways, ends the rejection. Removing five other models at random instead keeps the rejection in all 24 draws. A follow-up, chosen after seeing this result, isolates two pairs. Removing one model from Phi-3-mini/Phi-3.5-mini and one from Llama-3-8B/Llama-3.1-8B raises p_M above 0.2 under both nulls. None of 190 removals of two other models reaches 0.2. Llama-3.2-1B carries part of the excess: removing it alone ends the rejection ($p_M = 0.059$ for the 2PL, 0.089 for Rasch), though 12 of the 19 removals of it with another non-pair model still reject under the 2PL. Whether all-4-correct removes a preference that also tilts answers a model partly knows cannot be tested: the soft-preference null is not adequate there. Letter shares flag Llama-3.2-1B, Qwen2.5-0.5B and SmolLM2-1.7B, and 2PL residual correlations flag the two pairs (App. F).

4. Discussion

Within one model, detectable cross-question structure comes only from a shared seed or prompt; across models, the largest structure beyond skill is a letter preference, strongest in one model; after rotation, a marginally significant, fragile excess remains, tied post hoc to near-duplicate models. A letter preference, even one carried by a single model, can create an eigenvector that an ensemble would read as missing knowledge. Scores that treat cross-model disagreement as epistemic (Hamidieh et al., 2026) should be checked for letter preference. A shared seed leaves the distribution of each question's answers unchanged but feeds the same random numbers to every question in the pass, so any analysis that compares or pools answers across questions can detect it.

The excess is marginal (Sec. 3.3). The 217 questions with an unparseable reply may carry part of it, though dropping them also leaves the test less power. The adequacy rule, the single-model and pair removals that isolate the two pairs, and the analyses of test size, power and the difficulty factor were chosen after seeing the data (App. F). Why the models in a pair fail together was not tested; shared knowledge gaps from common training data, contamination and other causes remain open. In the ensemble, architecture, family, scale and training data vary together. The test can also miss structure carried by a few models, and anything outside the correctness matrix, such as log-probabilities. The within-model runs use instruction-tuned models of 1.7B to 8B, and the across-model runs 500 MMLU questions. No threshold is corrected for the nulls and scoring rules tried.

Before an ensemble's errors are read as missing knowledge, each answer needs its own seed. The options should be rotated, a question counted correct only if it is right in every rotation, and any remaining structure tested against a fitted one-skill model and checked for near-duplicate models.

References

Bai, Z. and Silverstein, J. W. *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics. Springer, New York, 2 edition, 2010. doi: 10.1007/978-1-4419-0661-8.

Bao, Z., Pan, G., and Zhou, W. Tracy–Widom law for the extreme eigenvalues of sample correlation matrices. *Electronic Journal of Probability*, 17:1–32, 2012. doi: 10.1214/EJP.v17-1962. arXiv:1110.5208.

Burnell, R., Hao, H., Conway, A. R. A., and Hernández-Orallo, J. Revealing the structure of language model capabilities. *arXiv preprint arXiv:2306.10062*, 2023.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Corvelo Benz, N. L., Tsirtsis, S., Straitouri, E., Chatzi, I., Artola Velasco, A., Thejaswi, S., and Gomez-Rodriguez, M. Evaluation of large language models via coupled token generation. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 300 of *PMLR*, pp. 2215–2223, 2026.

Ding, K. When LLMs agree, are they right? Auditing self-consistency and cross-model agreement as confidence signals. *arXiv preprint arXiv:2607.08065*, 2026.

Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024. doi: 10.1038/s41586-024-07421-0.

Fei, X., Zhang, Y., Carneiro, S. A., and Vazirgiannis, M. Every wrong answer counts: Option-level psychometrics for LLM multiple-choice benchmarks. *arXiv preprint arXiv:2608.02966*, 2026.

Ferguson, G. A. The factorial interpretation of test difficulty. *Psychometrika*, 6(5):323–329, 1941. doi: 10.1007/BF02288588.

Hamidieh, K., Thost, V., Gerych, W., Yurochkin, M., and Ghassemi, M. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations*, 2026.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.

Hüllermeier, E. and Waegeman, W. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. doi: 10.1007/s10994-021-05946-3.

Jiang, T. The limiting distributions of eigenvalues of sample correlation matrices. *Sankhyā*, 66(1):35–48, 2004.

Johnstone, I. M. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29(2): 295–327, 2001. doi: 10.1214/aos/1009210544.

Kendall, A. and Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Kim, E. M., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 30038–30066, 2025.

Kipnis, A., Voudouris, K., Schulze Buschoff, L. M., and Schulz, E. metabench – a sparse benchmark of reasoning and knowledge in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.

Lakshminarayanan, B., Pritzel, A., and Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., Chen, K., and Lin, D. MMBench: Is your multi-modal model an all-around player? In *Computer Vision – ECCV 2024*, volume 15064 of *Lecture Notes in Computer Science*, pp. 216–233. Springer, 2024. doi: 10.1007/978-3-031-72658-3_13.

Lord, F. M. and Novick, M. R. *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968. With contributions by Allan Birnbaum.

Maimon, A., Cohen, A. D. N., Vishne, G., Ravfogel, S., and Tsarfaty, R. From benchmarks to skills: Low-rank factors for LLM evaluation. *Transactions of the Association for Computational Linguistics*, 14:1639–1653, 2026. doi: 10.1162/tacl.a.745.

Manakul, P., Liusie, A., and Gales, M. J. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9004–9017, 2023. doi: 10.18653/v1/2023.emnlp-main.557.

Marčenko, V. A. and Pastur, L. A. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967. doi: 10.1070/SM1967v001n04ABEH001994.

McDonald, R. P. and Ahlawat, K. S. Difficulty factors in binary data. *British Journal of Mathematical and Statistical Psychology*, 27(1):82–99, 1974. doi: 10.1111/j.2044-8317.1974.tb00530.x.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11048–11064, 2022. doi: 10.18653/v1/2022.emnlp-main.759.

Pezeshkpour, P. and Hruschka, E. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 2006–2017, 2024. doi: 10.18653/v1/2024.findings-naacl.130.

Pillai, N. S. and Yin, J. Edge universality of correlation matrices. *Annals of Statistics*, 40(3):1737–1763, 2012. doi: 10.1214/12-AOS1022. arXiv:1112.2381.

Rasch, G. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, Copenhagen, 1960.

Ravfogel, S., Svete, A., Snæbjarnarson, V., and Cotterell, R. Gumbel counterfactual generation from language models. In *International Conference on Learning Representations*, 2025.

Ruan, Y., Maddison, C. J., and Hashimoto, T. Observational scaling laws and the predictability of language model performance. In *Advances in Neural Information Processing Systems*, 2024.

Sha, T. and Zhao, S. BenchScope: How many independent signals does your benchmark provide? *arXiv preprint arXiv:2603.29357*, 2026.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Yao, L. H., Jarvis, N., Zhan, T., Ghosh, S., Liu, L., and Jiang, T. A geometric lens on LLM abilities through joint embedding item response theory. *Transactions on Machine Learning Research*, 2026. URL https://openreview.net/forum?id=VQe6p9wn5g. arXiv:2509.22888.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 4791–4800, 2019. doi: 10.18653/v1/P19-1472.

Zhao, Z., Wallace, E., Feng, S., Klein, D., and Singh, S. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, pp. 12697–12706, 2021.

Zheng, C., Zhou, H., Meng, F., Zhou, J., and Huang, M. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, 2024.

A. Background

The eigenvalue test and the one-skill models of educational testing are classical. What is new is the effect of a letter preference on a correctness matrix.

Eigenvalue tests. n questions with pairwise correlation κ have a population eigenvalue of $1 + (n-1)\kappa$, which grows with n, so an eigenvalue test can detect a correlation smaller than the noise of one pairwise correlation, about $1/\sqrt{K}$. The MP law holds for sample covariance matrices of independent entries with finite variance (Marčenko & Pastur, 1967; Bai & Silverstein, 2010) and for sample correlation matrices of i.i.d. entries (Jiang, 2004). For Gaussian data, the largest eigenvalue minus λ_+, divided by $\sigma_{\mathrm{TW}} = (1+\sqrt{\gamma})(1+1/\sqrt{\gamma})^{1/3}K^{-2/3}$, converges to the Tracy–Widom law (Johnstone, 2001). That ratio is our z. For correlation matrices the limit is proved, as far as we know, only without centering or for Gaussian entries (Bao et al., 2012; Pillai & Yin, 2012). Our columns are Bernoulli, with unequal pass rates, and they are centered at their sample means, so z is calibrated by simulation (App. C).

Option position. Ding (2026) shuffle the options and find that part of a model's self-agreement depends on option position. In the option-level response model of Fei et al. (2026), a per-model positional preference barely helps to predict held-out correctness (accuracy 0.840 with it, 0.838 without). Sec. 3.2 shows that such a preference can still create the second eigenvector of a correctness matrix. We did not test whether item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026), or shared wrong answers (Kim et al., 2025), carry the same letter confound.

B. Data collection

Within one model, no count changes when unparseable replies are dropped or imputed. Across models, dropping every question with an unparseable reply ends the marginal all-4-correct rejection.

Questions. MMLU: ten random questions from each of the first 50 subjects in alphabetical order. HellaSwag: 200 validation items, sampled within each activity label. GSM8K: the first 100 test problems. In the shared-prompt runs, each pass draws its own five exemplars from the 285 MMLU development questions (so two passes can share one) and shows them with every question; both runs use the same sets.

Prompts and decoding. Each model receives one user message in its own chat template (raw text for Phi-2, which has none), checked against an independent reference tokenization; shared-prompt messages place the five exemplars, with their answer letters, before the question. Generation uses vLLM 0.23.0 in bfloat16, with plain temperature sampling and at most 8 new tokens for MMLU and HellaSwag, and 256 or 1024 for GSM8K. The 24-model runs decode greedily, with deterministic kernels. Every sample (k, i) has its own seed, except in the shared-seed run. vLLM gives each seeded request its own random-number generator, so the noise does not depend on the request's position in a batch.

Scoring. A multiple-choice reply counts as letter X in either of two cases. It begins with X, optionally after "Answer:", a parenthesis or bold markers, and it does not negate or hedge that letter. Or its first line says "the answer is X", or an equivalent phrase. A reply that names several letters, negates its letter, or fits neither case is unparseable and is scored wrong. A GSM8K reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. In the within-model runs, at most 1.65% of replies are unparseable (Llama-3-8B on Hel-

laSwag); in the ensemble, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%. Dropping every question with an unparseable reply, or imputing each such reply by the question's mean, leaves every count of the independent-draw runs at zero and both counts of Llama-3-8B's shared-prompt run at two. On all-4-correct, the max-statistic test still rejects after imputation, and after dropping the 95 questions whose score an unparseable reply decides. It does not reject after dropping all 217 questions that have one (Table A2). In rotation 0, dropping the 112 questions with an unparseable reply moves rank 2 to the 98.3rd percentile of the letter-aware null. Rank 2 stays above the 3PL null. The max-statistic test rejects the letter-aware null ($p_M = 0.013$) and the 3PL ($p_M = 0.004$; App. F).

Truncation on GSM8K. At 256 new tokens, 55% of Qwen2.5-7B's replies are cut off, mostly before the final answer (13.6% of them correct, against 96.7% of complete ones; accuracy 51.0% at 256 tokens and 90.3% at 1024). Truncation changes which answers are correct but does not couple two questions of a pass, so the run still tests independence.

C. Calibration of the within-model test

No theorem we know of gives the limit of z for our matrix (App. A), so z and the MP count are judged against Monte Carlo nulls of independent answers. Against those nulls, no independent-draw run shows structure at any borderline threshold, and every shared-seed or shared-prompt run shows it at all of them.

The null and z. The null treats each entry c_{ki}, given the model and the prompts, as a Bernoulli variable with $\Pr(c_{ki} = 1) = p_i$, independent across questions and passes; the borderline filter acts on each column alone, so the kept columns stay independent. At $K \leq 100$ the null distribution of z is shifted from the Tracy–Widom law: every z of the 12 independent-draw runs is negative, and their Monte Carlo nulls of z have means between -1.98 and -1.47. A z above every draw of its null, as in the three shared-seed and shared-prompt runs, rejects independence. The rejection does not say whether the dependence lies between questions or between passes.

Checks of independence. Three further checks agree with independence. The number of correct borderline answers per pass varies as independent draws predict: the ratio of observed to predicted variance runs from 0.61 to 1.09. No run is overdispersed against a column-shuffle null (smallest $p = 0.27$). The two lowest ratios, $p = 0.039$ and 0.045 for underdispersion, are not significant after a Holm correction over 48 tests, from 12 runs, over- and underdispersion, and two trends over the pass index. This check cannot detect a dependence that cancels in the number correct, such as the shared seed's (App. E). Borderline questions of one subject are no more correlated than other pairs (Qwen2.5-7B and Llama-3-8B, MMLU, $T = 1$: mean -0.018, $p = 0.80$, and 0.002, $p = 0.37$). No pair of borderline questions is coupled beyond its permutation null, in correctness for the five MMLU runs at $T = 1$ or in the answer letters of Qwen2.5-7B (largest Cramér's V 0.41, null 99th percentile 0.72).

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge; TW z: Tracy–Widom statistic. Counts: MP (above λ_+) / buffered (above $1.05\,\lambda_+$, a margin for the fluctuation of $\lambda_{\max}$). GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	TW z	**Counts**
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	-1.72	0/0
Mistral-7B-v0.3	83	3.65	3.45	-1.79	0/0
Llama-3-8B	173	5.36	4.97	-3.05	0/0
SmolLM2-1.7B	406	9.09	8.74	-2.17	0/0
Phi-3-mini	271	7.00	6.91	-0.65	0/0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	-1.60	0/0
Mistral-7B-v0.3	65	4.58	4.05	-2.71	0/0
Llama-3-8B	86	5.34	4.75	-2.90	0/0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	-2.06	0/0
GSM8K, 1024 tok.	19	3.22	2.50	-2.97	0/0
MMLU, $T = 0.5$	25	3.66	3.39	-1.08	0/0
MMLU, $T = 1.5$	85	7.20	6.53	-2.07	0/0
Shared seed or prompt, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	$+70.2$	4/4
Qwen, prompt	93	3.86	4.77	$+7.9$	2/2
Llama, prompt	264	6.89	11.31	$+30.9$	2/2

Borderline threshold. At $\varepsilon = 0.01, 0.02, 0.05, 0.10, 0.15$ and 0.20, with the filter re-applied to the data and to each of the 1000 null matrices, no independent-draw run has an eigenvalue above its edge or a z above the 95th percentile of its null, and every shared-seed or shared-prompt run keeps at least two eigenvalues above its edge. The six tests of a run are nested, and at $\varepsilon = 0.20$ the three smallest sets have only 6, 7 and 10 questions.

D. Power of the within-model test

The within-model test misses some weak structure. A factor shared by all borderline questions of a pass, with latent correlation 0.05, is found in every simulation at 173 borderline questions but in only 47% at 43. A cluster of nine questions is usually missed at every strength we simulated.

Simulation. The matrices have the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$), $K = 100$ passes and all 500

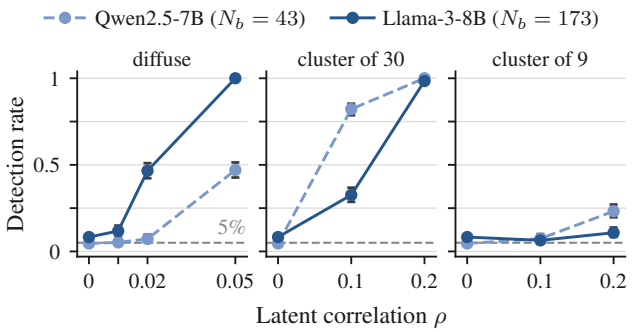


Figure A1. Detection rate of the MP count against the latent correlation ρ of a factor planted on all borderline questions or on 30 or 9 of them, at the pass rates of two MMLU runs. Whiskers: Wilson 95% intervals.

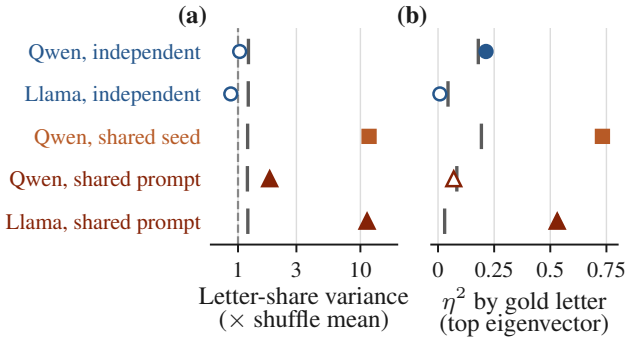


Figure A2. A shared seed or prompt makes a pass's letter shares vary beyond chance. Of the shared runs, only the seed run and Llama-3-8B's prompt run also group the top eigenvector's loadings by gold letter. **(a)** Across-pass variance of a pass's letter shares, summed over letters, divided by the mean of that sum after answers are shuffled across passes. **(b)** η^2 of the top eigenvector's loadings by gold letter. Grey tick: null 99th (a) or 95th (b) percentile; filled: beyond it.

questions. Entry c_{ki} is 1 when a hidden standard normal score falls below $\Phi^{-1}(f_i)$, so each question keeps its pass rate. On loaded questions the score is $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$, with u_k shared by the loaded questions of pass k and e_{ki} independent noise; elsewhere it is e_{ki}. Two loaded questions then correlate at ρ in their scores and, at these pass rates and averaged over pairs, at about $\rho/2$ in their 0/1 entries. The factor loads on all borderline questions (diffuse) or on 30 or 9 chosen at random. A count of one or more, with the filter re-applied, detects the factor. Each scenario has 500 matrices (2000 at $\rho = 0$).

Results. A diffuse factor is easier to detect at Llama-3-8B's 173 borderline questions, because it loads four times as many (Fig. A1). A cluster of fixed size is harder to detect there, because it is a smaller share of them: a 9-question cluster at $\rho = 0.2$ is detected in 23% of Qwen2.5-7B and 11% of Llama-3-8B simulations. The false-positive rates are 4.6% and 8.3% for the MP count and 0.65% and 0.35% for the buffered count. The buffered count has less power: 22% and 99.8% for a diffuse factor at $\rho = 0.05$. We did not simulate other values of N_b.

E. What a shared seed or prompt couples

A shared seed couples the questions through the sampler's noise on each letter. A shared prompt couples them through the exemplars' answer letters for Llama-3-8B, and through the exemplar set itself for Qwen2.5-7B.

Shared noise. At each generation step, a seeded request draws one exponential random number for every token. It emits the token whose probability at temperature T, divided by that number, is largest. This is Gumbel-max sampling, so each answer on its own follows the model's distribution. Two requests with the same seed draw the same numbers at every step, whatever their prompts. Coupled generation and counterfactual sampling use that sharing on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025). If a shared-seed pass draws a small number for "B" at some step, "B" is favored in every question whose answer letter is generated at that step.

The shared-seed run. In Qwen2.5-7B's shared-seed run, the across-pass variance of a pass's letter shares (its fractions of A, B, C and D answers), summed over letters, is 11.8 times the mean of that sum after each question's answers are shuffled across passes, against 1.03 in its independent-draw run (Fig. A2a). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others, so the top eigenvector's loadings group by gold letter (Sec. 3.1), while the gains and losses cancel in the number correct per pass (observed over predicted variance 0.83).

The shared-prompt runs. Llama-3-8B copies the exemplars' letters. Few-shot prompts are known to induce that copying (Zhao et al., 2021). The number of a pass's exemplars whose answer is letter L correlates with its share of answers L at $r = 0.35$ to 0.47. Its letter shares vary 11.3 times their shuffle mean, and its number correct per pass is overdispersed (ratio 1.51, $p = 0.001$). Qwen2.5-7B's letter shares vary 1.8 times their shuffle mean and do not follow the exemplars' letters ($|r| \leq 0.09$; loadings by gold letter $\eta^2 = 0.07$). Its passes that share an exemplar still agree more on the borderline questions than passes that share none, by 10.7 standard deviations of a null that permutes exemplar sets (5.9 standard deviations for Llama-3-8B; $p = 0.0005$ for both). Links to the exemplars' content are weak. Subject grouping of the loadings has $p = 0.042$, uncorrected, and a same-subject exemplar adds 3.2 points of accuracy on borderline questions ($p = 0.033$). Neither Llama-3-8B's top eigenvector nor that of the shared-seed run groups by

subject ($p = 0.35$ and 0.23).

Gold letter below the edge. Qwen2.5-7B's independent-draw run shows $\eta^2 = 0.21$ by gold letter ($p = 0.023$; Fig. A2b), although its $\lambda_{\max}$ lies below λ_+ and the answer-letter test of App. C finds no coupled pair. That test looks for single coupled pairs, not for a grouping of loadings. We do not interpret this value.

F. The 24-model ensemble

In rotation 0, every adequate null whose letter preference, if it has one, acts only on guesses leaves the second eigenvalue in excess. The soft-preference null, whose preference acts on every answer, makes that eigenvalue unlikely and still fits the data poorly. On all-4-correct, the marginal excess ends when one model from each of two pairs is removed, when Llama-3.2-1B is removed ($p_M = 0.059$ and 0.089), or when the 217 questions with an unparseable reply are removed.

F.1. Models, nulls and the max-statistic test

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B; Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-2 (2B, 9B), and SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B. Phi-2 is a base model; the other 23 are instruction-tuned. Rotation-0 accuracy runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-skill nulls. In every null, skill is the only link between questions, and every simulated draw is conditioned on each kept column staying split. The 2PL (Sec. 2) treats the skills as fixed and integrates the question parameters over a normal population, by marginal maximum likelihood. Each simulated column draws its (a_i, b_i) from the posterior fitted on all 500 columns. Rasch is fitted by joint maximum likelihood. The 3PL is $0.25 + 0.75\,\sigma(a_i(\theta_m - b_i))$. In the uniform know-or-guess null (KG-u), a model either knows a question, with the 2PL probability, in all four rotations at once, or guesses a letter uniformly in each rotation. KG-u is fitted to all four rotations and scored like the data. The letter know-or-guess null (KG-l) guesses from the model's letter preference π_m instead. The letter-aware null is fitted to rotation 0 only, with π_m taken from all four rotations. The soft-preference null is defined in App. F.2. We also tried a Bock–Aitkin 2PL, with skills integrated and slopes at most 4, and a fixed-margin null, uniform over 0/1 matrices with the observed margins.

Adequacy. A null counts only if the observed λ_1 lies between its 1st and 99th percentiles. In rotation 0 (1000 draws per null) it lies at the 76.4th percentile of the 3PL, 86.7th of the letter-aware, 98.7th of the KG-u and 85.7th of the SP null, which are adequate, and at the 100th of the 2PL and Rasch. On all-4-correct (2000 draws) it lies at the 31.5th and 54.2nd percentiles of the 2PL and Rasch, which are adequate, at the 100th of the 3PL and at the 0th of the KG-u and SP. KG-l, Bock–Aitkin and fixed-margin are inadequate for both. An adequate null's per-rank count is the number of ranks $2, 3, \ldots$ that each exceed its 99th percentile, counted until one does not; the rank-2 rule asks only about rank 2.

Max-statistic test. The test covers ranks 2 to 4. It uses 2000 new null draws, with seeds disjoint from all earlier runs. Its statistic M is the largest of the percentiles of λ_2, λ_3 and λ_4 among these draws, ties counted as half. Each draw's own M_d is computed against the other draws, and $p_M = (1 + \#\{d : M_d \geq M\})/2001$, at level 0.05. Three null draws reach $M = 100$, so p_M cannot fall below 0.002, the value in rotation 0.

Order of analyses. The adequacy rule was set after the first analyses. The max-statistic test and the list of sibling groups were written down before that test was run. The tests are still not independent of what came earlier: a statistic that pools ranks 2 to 4 had already rejected on these data, and the separate rank tests and the correlations between model pairs had already been computed. Removing single models or pairs, and the analyses of test size, power and the difficulty factor, were decided after seeing the results.

Size. From each fitted one-skill model, 2PL and Rasch, we simulate 300 all-4-correct datasets, refit both nulls to each (500 draws), and apply the adequacy rule and the max-statistic test (Table A2). The 2PL test rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (95% intervals 4.1–10.1% and 5.7–12.4%), so it is not size-controlled by the bound of 7.5% we set before the run. The Rasch test rarely rejects under Rasch truth (0.7%), mainly because the refitted Rasch null is adequate in only 23% of those datasets. The adequacy rule is thus not sufficient: it accepts the Rasch null on every 2PL dataset, despite wrong slopes, and rejects the correct Rasch model in most datasets drawn from it.

Other scores. A question can also count as correct in at least one, two or three rotations, or by the mean of the four. The 2PL rejects at "at least two" ($p_M = 0.002$). Rasch, KG-l and the 2PL reject at "at least three" ($p_M = 0.008$, 0.007 and 0.009). The 3PL and the soft-preference null do not, at "at least two" ($p_M = 0.86$ and 0.85), nor does KG-u at "at least three" ($p_M = 0.28$). Datasets drawn from

the soft-preference fit, which has no second factor, make a refitted 2PL reject in 98.5% of datasets at "at least two" and in 58% at "at least three", so these scores support no conclusion. "At least one of four" gives $p_M = 0.996$. The mean score has no adequate null.

F.2. Rotation 0: the letter eigenvector

Nulls with a letter preference. The letter-aware, 3PL and KG-u nulls let a letter preference act only on guesses; the SP null lets it also tilt answers a model partly knows. In 200 datasets simulated from the SP fit, which has no second factor, rank 2 exceeds the letter-aware null in 60% of datasets and the 3PL null in 99.5%. Each null is refitted, with 200 draws. A letter preference of this kind, with no second skill, is often enough to put rank 2 above those two nulls.

The soft-preference null. Model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$, where g_i is the correct option, b_{ij} is a fitted intercept for option j with $b_{ig_i} = 0$, L_j is the letter at which the rotation places option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of one model on one question and independent across models. The weight $a_i\theta_m + e_{mi}$ is added only for the correct option. π_m multiplies every option, and skill is the only link between questions. The fit is penalized maximum likelihood on the parsed choices of all four rotations ($\tau = 1.17$). Simulations drop the cells that are unparseable in the data. Without that mask, the observed λ_2 lies at the 99.8th percentile of the null; with the mask, at the 95.1st. Refitting inflates the slopes, so the tail probabilities are recalibrated on 200 datasets drawn from the fit, each with its own null of 1000 draws. The calibrated p of rank j is the fraction of those datasets whose percentile for λ_j is at least the observed percentile. In rotation 0 that p for λ_2 is 0.010, or 0.015 under the convention $(1 + k)/(1 + n)$. The max-statistic test does not reject ($p_M = 0.079$). The null is inadequate without Llama-3.2-1B and on all-4-correct, so a preference that tilts partly known answers is untested under all-4-correct.

Predictive check. The SP model fails its predictive check, set before the run, against 500 datasets simulated from the fit. Per-model accuracy and letter shares fit, but the number of rotations in which a model answers a question correctly is covered in only 82.5% of model-by-count cells (90% required), and the observed number of split questions under "correct in at least two rotations" lies above all simulations. The weakest, letter-biased models have too many questions correct in exactly one rotation, and 21 of 24 models too many in all four (largest excess 0.06). The calibration is exact only if each refit inflates the slopes as much as the first. It does not correct the misfit, so the soft-preference verdicts are not fully validated.

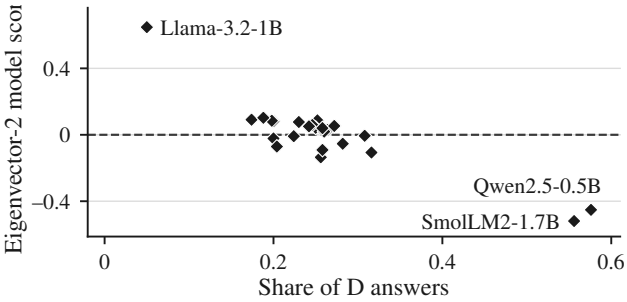


Figure A3. Each model's rotation-0 score on eigenvector 2 of the 24-model matrix against its share of D answers.

Letter evidence. The models' scores on eigenvector 2 track their share of D answers ($r = -0.90$, permutation $p = 10^{-4}$; Fig. A3). Eigenvector 1 also carries a letter part ($\eta^2 = 0.22$ by gold letter), and eigenvectors 3 and 4 group by gold letter more weakly, though still beyond chance ($\eta^2 = 0.087$ and 0.061, $p = 0.0002$ each). Because gold letters are nearly balanced (118 to 129 per letter), a letter preference leaves expected benchmark scores nearly unchanged.

Without Llama-3.2-1B. The 3PL is still rejected ($p_M = 0.003$), and so is the letter-aware null ($p_M = 0.015$). The letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$). Ranks 3 and 4 lie at the 98.5th to 99.9th percentiles of these two nulls, and in their 1000 draws rank 4 exceeds the 99th percentiles by only 0.35 and 0.57.

F.3. All-4-correct: the marginal excess

Power. We simulate from the adequate all-4-correct 2PL fit plus a second skill. As in the 2PL null, each of the 391 kept columns draws (a_i, b_i) from its posterior. On the 125 of them from the 18 STEM subjects, the logit gains $s\,\bar{a}\,\mathrm{sd}(\theta)\,z_m$, where $z_m \sim N(0, 1)$ is independent of θ_m, $\bar{a} = 1.71$ is the median slope and $\mathrm{sd}(\theta) = 1$. At $s = 1$ the second skill matches the first in spread and in median slope. We refit both nulls to each of 200 datasets per s (500 draws; rates in Table A2). At $s = 1$ a test of λ_2 alone detects the second skill in 69 to 78% of datasets. At $s = 2$ the max-statistic test and the rank-2 rule detect the second skill in at least 97% of datasets. The median λ_2 is 28.2 at $s = 0$, 28.3 at $s = 0.5$ and 33.8 at $s = 1$, against the observed 29.0.

The difficulty factor. In the all-4-correct nulls, the eigenvector-2 loadings follow question difficulty: the observed $|r|$ between loading and difficulty, 0.57, lies just below the 2PL null's 5th percentile and inside the Rasch null's 5–95% range. In the rotation-0 3PL null, by contrast, the median $|r|$ is 0.09 (observed 0.04). The observed load-

Table A2. Max-statistic test on all-4-correct (reject at $p_M < 0.05$). Subsets: number that reject, and the range or maximum of p_M. Pair removals: one Phi and one Llama model. Other two removed: the 190 removals of two other models (fraction that reject, maximum p_M). Unparseable: replies imputed, or questions dropped, either the 95 whose score they affect or all 217 that contain one (1000 draws). Size: rejection rate in 300 datasets simulated from each fitted one-skill model. Calibrated p_M: under 2PL and Rasch truth. Power: max-statistic / rank-2 rule, 200 datasets per strength s (2PL fit).

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_M, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Other two removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_M	0.033, 0.047	0.057, 0.017
Power, $s = 0.5$	0.20 / 0.01	0.21 / 0.07
Power, $s = 1$	0.91 / 0.69	0.93 / 0.78

ings group by subject ($\eta^2 = 0.21$, $p = 0.001$). A cubic in difficulty explains 39% of their variance, and once that trend is removed the subject grouping is at chance ($p = 0.24$).

Model families. On all-4-correct, the models' scores on eigenvector 4, whose eigenvalue lies below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$). In rotation 0, same-family pairs correlate little more than other pairs (mean r 0.414 against 0.375, $p = 0.085$).

Sibling models. Keeping one model from each sibling group (Sec. 3.3) gives $2 \cdot 2 \cdot 3 \cdot 2 = 24$ subsets of 19 models. None rejects. Both nulls are refitted, with 1000 draws, and both are adequate in every subset. All 24 controls reject. Each removes five of the other 15 models at random (Table A2). The subsets overlap, so this is one contrast, not 24 replicates. The five removed models are not matched in size to the sibling models, and the controls include other models from the same developers (Phi-2, Qwen2.5 at 0.5B to 3B, Llama-3.2, Ministral-8B, Mistral-Small). In rotation 0 the 3PL and letter-aware nulls are adequate, and both reject, in all 24 subsets.

The two pairs. Removing one model from each of the two pairs in Sec. 3.3, in all four ways, gives a larger p_M under both nulls than any of the 190 removals of two other models. Of those 190, 90% still reject under the 2PL and 74% under Rasch (1000 draws; every null adequate; Table A2). The Qwen pair is not part of the excess: removals that include Qwen2-7B or Qwen2.5-7B reject more strongly, with mean p_M 0.010 and 0.006 under the 2PL. Nor is the Mistral group: removing Mistral-7B v0.2 and v0.3, the third-ranked pair in Table A3, leaves $p_M = 0.019$, and the other two removals in that group give p_M from 0.004 to 0.023. Nor is accuracy: the 15 removals of two models with rotation-0 accuracy 0.62 to 0.71, around the pair models' 0.644 to 0.684, give p_M at most 0.052 under the 2PL and 0.113 under Rasch, and every pair removal has a larger p_M.

Table A3. Simple baselines. Top: rotation-0 letter shares of the three models whose largest share exceeds 0.5; no other model's largest share exceeds 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

Llama-3.2-1B. Removing Llama-3.2-1B alone gives $p_M = 0.059$ under the 2PL and 0.089 under Rasch, and it makes the KG-u null adequate ($p_M = 0.15$). The sum $\lambda_2 + \lambda_3 + \lambda_4$, fixed with the test and not used for decisions, still gives $p = 0.023$ and 0.016. Of the 19 removals among the 190 that include Llama-3.2-1B, 12 still reject under the 2PL and 6 under Rasch. Removing it together with one model from each pair gives p_M from 0.30 to 0.59 under the 2PL and from 0.38 to 0.60 under Rasch.

Simple baselines. Rotation-0 letter shares flag three models whose most frequent letter exceeds half of their replies (Table A3). These shares count unparseable replies. The preference of 0.90 for A in Sec. 1 is estimated over guesses. Residual correlations after the all-4-correct 2PL fit rank the Phi and Llama pairs first and second of 276.

F.4. Replicates, near ties and option references

Replicates. The ensemble was collected twice, to test GPU nondeterminism. The paper reports replicate A. Replicate B differs in 6 of 48,000 chosen options, and in eigenvalues by at most 0.081. In B alone, the all-4-correct λ_4 is 20.82, just above the 2PL rank-4 99th percentile of 20.78 from 1000 draws. A near tie is a cell whose two most probable first-token letters differ by less than 0.13 nats in either replicate (5.0% of cells). Replacing each near tie by

the majority correctness over the four rotations changes no verdict. Dropping the 297 questions with a rotation-0 near tie leaves 203. Rank 2 then exceeds the 3PL but not the letter-aware null, and both nulls are adequate. Removing the 23 questions whose options refer to other options changes no per-rank count in rotation 0 or on all-4-correct.